

2025-2026学年第二学期 《遥感与GIS综合》 期末考试试题

- 考试卷型：A卷、开卷
- 教师：丁凤 福建师范大学地理科学学院
- 课程：《遥感与GIS综合》
- 考试时间：2026年6月27日 14:00-16:00

本份试卷为回忆版试题。@Xuuyuan

I. Write the corresponding Chinese/English words for the following: (32%)

1. Census data
2. spatial interpolation
3. metadata
4. accuracy assessment
5. object-oriented classification
6. data redundancy
7. 特征提取
8. 光谱混淆
9. 人口估算
10. 不透水面
11. 误差分析
12. 数据融合
13. 变化检测
14. 辅助数据
15. 比例图像（组分图像）
16. 逐步回归分析

II. True or False: (10%)

1. Some of the newer satellite images have a spatial resolution of 0.5 meter or less.
2. Shapefiles are NOT topology-based.

3. An image with a larger pixel size has a higher resolution than an image with a smaller pixel size.
4. Digital orthophotos do not contain image displacement or distortion.
5. $MNDWI = (Green - NIR) / (Green + NIR)$.

III. Single choice questions: (10%)

1. Which of the following satellites has a panchromatic resolution of 0.8m and a multispectral resolution of 3.2m?
 - a. GF-2
 - b. Sentinel-2
 - c. IKONOS-2
 - d. MODIS
 - e. LJ1-01
 - f. Landsat-9
2. Which of the following statistics CANNOT be used in a neighborhood operation?
 - a. maximum
 - b. mean
 - c. majority
 - d. minority
 - e. none of the above
3. What was the name of the first artificial satellite?
 - a. Landsat-1
 - b. GF-1
 - c. Sputnik 1
 - d. SPOT-1
 - e. WorldView-1
4. Which of the following represents a secondary data source?
 - a. GPS readings
 - b. satellite images
 - c. aerial photographs
 - d. paper maps
5. The accuracy of spatial interpolation can be influenced by:
 - a. number of control points
 - b. distribution of control points
 - c. data-poor areas
 - d. all of the above

IV. Answer the following questions: (48%)

1. Ancillary data (including GIS data such as soil, terrain, zoning, and Census data, etc.) has long been used to improve the remote sensing image classification accuracy. Please **briefly describe** how the ancillary data can be used at different stages of image classification. You may **draw a flowchart** to facilitate your description. (10%)
2. According to Chapter 10 of the textbook, please briefly describe how to carry out population estimation at the country level using remote sensing data combined with Census data. You may **draw a flowchart** to aid your description. You are encouraged to **put forward suggestions for future improvement** to the research outlined in Chapter 10. (14%)
3. With the development of remote sensing technologies, tools and datasets, does this eliminate the need for people to collect data or information on the ground? Please **briefly explain why or why not**. (9%)
4. Based on what you have learned from the course of Remote Sensing and GIS Integration, what **critical considerations** should be kept in mind **when calculating the RSEI** (Remote Sensing Based Ecological Index)? Additionally, **what aspects require special attention** if we aim to **improve the RSEI** ? (15%)